Chapter 1: Introduction to Machine Learning

Machine learning is a subset of artificial intelligence that enables computers to learn without being explicitly programmed. It focuses on developing algorithms that can learn from and make predictions on data.

Chapter 2: Supervised Learning

Supervised learning uses labeled training data to learn a mapping function from inputs to outputs. Common algorithms include linear regression, logistic regression, decision trees, and support vector machines.

Chapter 3: Unsupervised Learning

Unsupervised learning finds hidden patterns in unlabeled data. Clustering algorithms like k-means group similar data points together. Dimensionality reduction techniques like PCA simplify data while preserving important information.

Chapter 4: Deep Learning

Deep learning uses neural networks with multiple layers to model complex patterns. Convolutional neural networks excel at image recognition. Recurrent neural networks process sequential data like text and time series.

Chapter 5: Natural Language Processing

NLP combines linguistics and machine learning to process human language. Transformers like BERT and GPT have revolutionized text understanding. Applications include translation, summarization, and question answering.

Chapter 6: Model Evaluation

Evaluating model performance is crucial. Metrics include accuracy, precision, recall, F1-score for classification. Cross-validation helps ensure models generalize well to unseen data.

Chapter 7: Deployment and MLOps

Deploying models requires monitoring, versioning, and automation. Tools like MLflow track experiments. Docker containers ensure consistency. CI/CD pipelines automate testing and deployment.